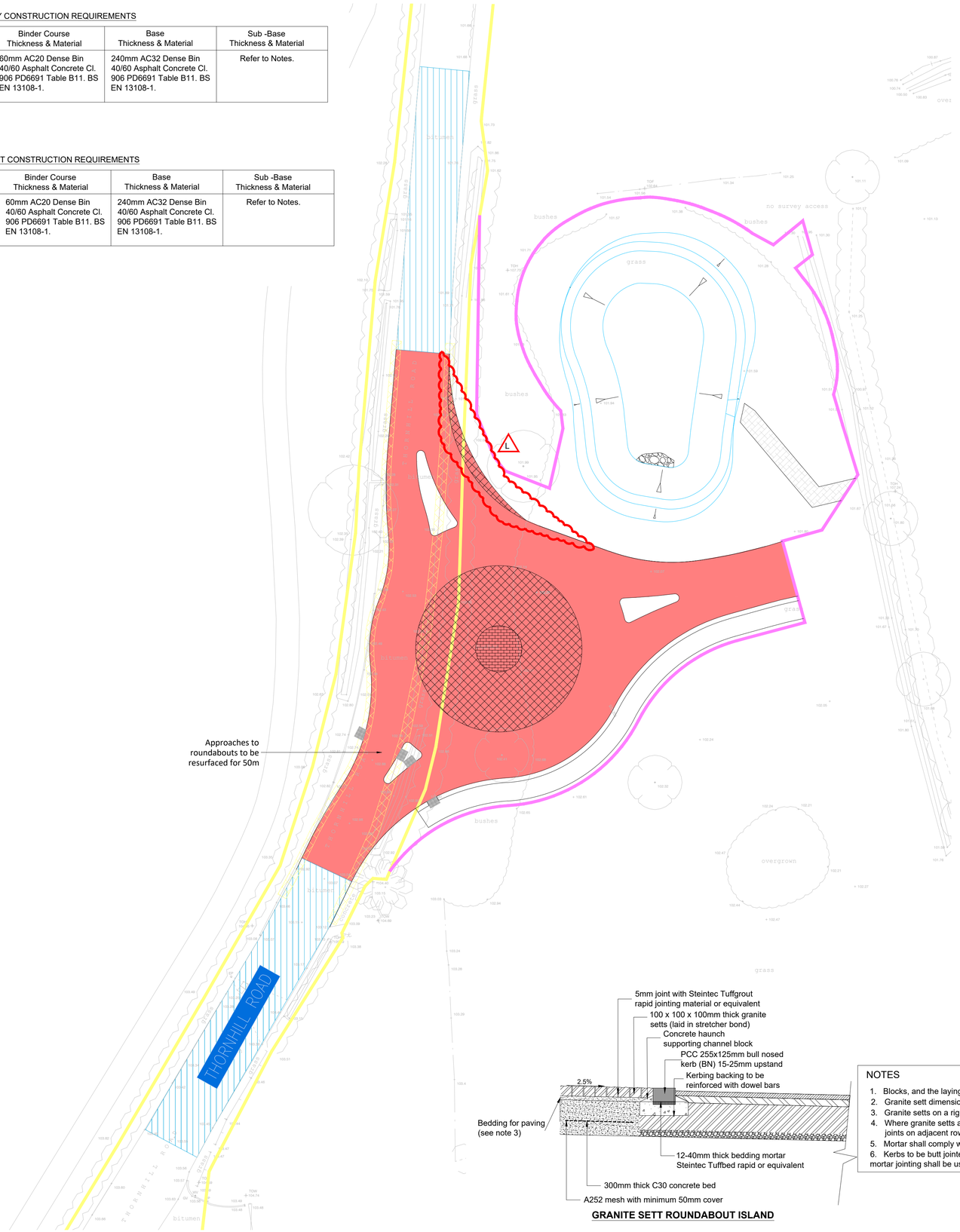
Original Drawing Size A1

CARRIAGEWAY CONSTRUCTION REQUIREMENTS

Road Type	Surface Course Thickness & Material	Binder Course Thickness & Material	Base Thickness & Material	Sub -Base Thickness & Material
	40mm Ultilayer PMB thin surface course (or similar approved) in accordance with Clause 908 and 942, to BS EN 13108-5 (PSV 68 AAV 16)	60mm AC20 Dense Bin 40/60 Asphalt Concrete Cl. 906 PD6691 Table B11. BS EN 13108-1.	240mm AC32 Dense Bin 40/60 Asphalt Concrete Cl. 906 PD6691 Table B11. BS EN 13108-1.	Refer to Notes.

ROUNDABOUT CONSTRUCTION REQUIREMENTS

Road Type	Surface Course Thickness & Material	Binder Course Thickness & Material	Base Thickness & Material	Sub -Base Thickness & Material
	40mm Ultilayer PMB thin surface course (or similar approved) in accordance with Clause 908 and 942, to BS EN 13108-5 (PSV 60 AAV 16)	60mm AC20 Dense Bin 40/60 Asphalt Concrete Cl. 906 PD6691 Table B11. BS EN 13108-1.	240mm AC32 Dense Bin 40/60 Asphalt Concrete Cl. 906 PD6691 Table B11. BS EN 13108-1.	Refer to Notes.



Construction Design and Management (CDM)
Key Residual Risks

Contractors entering the site should gain permission from the relevant land owners and/or principle contractor working on site at the time of entry. Contractors shall be responsible for carrying out their own risk assessments and for liaising with the relevant services companies and authorities. Listed below are Site Specific key risks associated with the project.

1) Overhead and underground services
2) Street Lighting Cables
3) Working adjacent to water courses and flood plain
4) Soft ground conditions
5) Working adjacent to live highways
6) Unchartered services

KEY:

- Existing Highway Boundary
- Proposed S278 Boundary
- Full carriageway construction
- Cold mill existing surface course and relay surface course with fine regulating material subject to site investigation
- Stepped tie into existing pavement
- Grasscrete
- Granite Setts on Concrete Base
- Ashphalt Block Paving
- Potential future pedestrian crossing facility

Foundation CBR Table

Pavement foundations are to be designed as part of the technical submission made to SBC, the foundation is to be based on the lowest CBR result from the ground investigation (GI) report. If the GI report doesn't include CBR results then a 2% CBR is to be assumed until insitu CBR testing is undertaken. *Swindon Borough is predominantly Kimmeridge clay which is very moisture and seasonally sensitive.*

CBR testing is required at 25m centres, the lowest CBR result is to be used to determine the needed foundation. Foundation requirements are to be approved by SBC's Clerk of Works/Engineer before the foundation is constructed, this will require insitu CBR testing in accordance with BS1377-9: 1990 Clauses 4.1 & 4.3.

CBR (%)	Subbase on Capping (mm)		Subbase only (mm)
	Subbase	Capping	
≤ 2.4	250	420 + Ground Stabilisation See ground stabilisation options below	
2.5 – 2.9	250	420	
3.0 – 3.9	240	380	
4.0 – 4.9	220	320	N/A
5.0 – 5.9	200	260	300
6.0 – 7.9	180	240	300
8.0 – 9.9	160	230	250
10 – 14.9	150	200	230
≥15.0	150	150	200

(Figures used in the table above have been extracted from figures 3.20 of CD 225 Rev 1 and figure 3.1 of HD 25/95)

Subbase only foundation are only permitted on CBRs of ≥5% (50MPa) in accordance with CD225 Clause 3.12.

Foundations on cohesive soils (e.g. clay) shall use subbase on capping foundation type.

All subbase shall be DOT Type 1 in compliance with MCHW1 803.

All capping shall be 6F5 in compliance with MCHW1 613.Subbase shall achieve a minimum CBR of 30% and capping shall achieve a minimum of 15%.

Grading certificates for all granular fill shall be provided for every 500 tonnes.

Bond coat shall be applied as per MCHW cl.903/6. Bond coat shall be applied at a rate of 0.2kg/m2 on newly laid binder/base courses and 0.35kg/m2 on planed surfaces by a metered spray tanker. Bond coat shall comply with BS EN 13808.

Ground Stabilisation

CBR results of less than 2.5% (30MPa) require ground stabilisation in addition to the foundation from the above table. The method of ground stabilisation and design is to be agreed with SBC's Clerk of Works/Engineer prior to implementation.

Implementation without SBC's Clerk of Works/Engineer approval and inspection could result in the road becoming unadoptable or requiring remediation works at the contractors/developers expense.

There are 2 permitted ground stabilisation methods available, these are:

Increased capping – if a suitable load bearing subgrade is within 1.0m of the foundations formation the soft subgrade is to be excavated and replaced with 0.5m-1.0m 6F5, in accordance with CD225 Clause 2.7 Note 2.

Geo-grid – is to only be used where there are no utilities/services beneath the pavement. SBC will not accept this method post technical approval and without evidencing that all other alternative methods weren't possible. All geo-grid proposals must have a comprehensive design and risk assessment.

Construction on Made Ground

Due to the unknown and high risk nature of made ground SBC requires made ground to be removed prior to construction of all highway works.

Bound Materials to be Removed

All bound materials and concrete elements remaining from existing carriageway and footway areas that are located under proposed verge or landscaped areas in the final layout are to be fully removed and disposed off off-site.

GRANITE SETT ROUNDABOUT ISLAND

NOTES

1. Blocks, and the laying of blocks shall comply with S.H.W. Clause 1107.
2. Granite sett dimensions shall be 210mm x 110mm x 100mm
3. Granite setts on a rigid concrete base, bedded on Steintec Tuffbed mortar.
4. Where granite setts are laid in stretcher bond on curves, cut setts shall be inserted where necessary so that joints on adjacent rows are no closer together than one quarter of a block length.
5. Mortar shall comply with S.H.W. Clause 2402 designation (i).
6. Kerbs to be butt jointed. Where kerb alignments incur 'minor open joint tapers' between formed surfaces mortar jointing shall be used as directed by Highway Engineer.

NOTES:

1. Do not scale from this drawing.
2. All dimensions are in metres unless otherwise stated.
3. Brookbanks consulting ltd has prepared this drawing for the sole use of the client. the drawing may not be relied upon by any other party without the express agreement of the client and Brookbanks consulting ltd. where any data supplied by the client or from other sources has been used, it has been assumed that the information is correct. no responsibility can be accepted by Brookbanks consulting ltd for inaccuracies in the data supplied by any other party. the drawing has been produced based on the assumption that all relevant information has been supplied by those bodies from whom it was requested.
4. No part of this drawing may be copied or duplicated without the express permission of Brookbanks consulting ltd.
5. All dimensions and all levels in metres unless noted otherwise.
6. Any discrepancies noted on site are to be reported to the engineer immediately.
7. Section 278 agreement highway works to carried out in strict accordance with Swindon Borough council's requirements.
8. All works within existing public highway, including any temporary works or traffic management measures, are subject to the approval of Swindon Borough Council. when works are required on the public highway, the contractor shall liaise with and obtain all statutory approvals from Swindon Borough Council, before commencing these works. these approvals include, but are not limited to, approval of traffic management measures, issue of works commencement notices, road opening notices, temporary traffic regulation orders etc.
9. All traffic management shall comply with the requirements as set out in chapter 8 of the traffic signs manual. warning signs may be erected outside the indicated boundaries. any obstructions to traffic or pedestrians shall be properly signed and protected with barriers, cones, signs, and lamps.
10. All planting in visibility splays to be agreed and approved by Swindon Borough Council and in all cases no planting to be above 600mm in height above the carriageway. also no obstructions of any kind within the visibility splay areas.
11. All tactile areas to be laid in accordance with detr "guidance on the use of tactile paving surfaces".
12. All traffic signs and road markings are to be in accordance with the "Traffic Signs Regulations and General Directions 2016'.
13. Any damage caused to existing footways, roads or other third party property to be made good to the county engineers requirements.
14. Temporary works associated with the construction of the works shall be the responsibility of the contractor.
15. The Highway Agreement's Clerk of Works is to be present during construction and CBR testing for inspections.
16. Pavement designs for all new/ widened carriageway areas should be fully in accordance with DMRB standards with a minimum required pavement design life of 40 years.
17. Pavement maintenance designs should be undertaken in accordance with the DMRB for all existing pavement areas within the permanent works, to achieve a renewal of the pavement life to match the 40-year design life of the new pavement areas.
18. Tar investigations and management should be undertaken in accordance with current good practice.

L	Updated following SBC TA4 comment 24/10/23	CM	CM	SM	03.11.23
K	Updated following SBC TA3 comment 12/10	CDW	CM	SM	18.10.23
J	Updated following SBC TA2 comment 26/09	CDW	CM	SM	06.10.23
H	Updated following TA1 comment	CM	CM	SM	26.09.23
G	Updated following TA1 comment	CM	CM	SM	24.08.23
F	Notes updated and stepped tie-in added	CDW	CM	SM	05.01.23
E	Updated following RSA comment	CM	SM	SM	20.09.22
D	Updated for SBC submission	CM	SM	SM	01.08.22
C	Updated to comments received from SBC June 2022	CDW	SM	RM	01.07.22
B	Footpath North of Roundabout Removed	CM	SM	RM	20.05.22
A	Updated S278 Boundary & Drainage Design	CDW	SM	RM	24.02.22
-	First Issue	PS	DS	PAB	30.11.21

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Swindon EDAG

Swindon EDA

Thornhill Road
Road Pavement Layout
SBC TA S1055

Status		Status Date
Approval		NOV' 2022
Drawn	**Checked**	**Date**
PS	DS	30.11.21
Scale	**Number**	**Rev**
1:250	1335/HL/56/700	L

0 5 10 15 20 25
METRES

UNTIL TECHNICAL APPROVAL HAS BEEN OBTAINED FROM THE RELEVANT LOCAL AUTHORITIES, IT SHOULD BE UNDERSTOOD THAT ALL DRAWINGS ARE ISSUED AS PRELIMINARY AND NOT FOR CONSTRUCTION. SHOULD THE CONTRACTOR COMMENCE SITE WORK PRIOR TO APPROVAL BEING GIVEN, IT IS ENTIRELY AT HIS OWN RISK.

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